

Fixed Income Securities

Question 1 will be covered in the Tutorial. Solutions to Questions 2-4 will be posted on Amathuba.

QUESTION 1

[22 marks]

Below you are given the Snapshot of a bond issued by the Republic of South Africa. Study the snapshot and answer the questions that follow.

SAGB 10 1/2 12/21/26 #18.580 +.048 8.598 /8.562

As of 12 A-- x--Source B

SAGB 10 1/2 12/21/26 #FActionsSettingsPage 1/12 Security Description: Bond

No NotesBuySell

29 Bond Description26 Issuer Description

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Issuer Information

NameREPUBLIC OF SOUTH AFRICAIndustryTreasury (BCLASS)Security InformationMkt IssDOMESTICCtry/RegZACurrencyZARRankSr UnsecuredSeriesR186Coupon10.500000TypeFixedCpn FreqS/ADay CntACT/365Iss Price77.36654Maturity12/21/2026BULLETIss Sprd

Calc Type(368)S AFRICA EX-DIV BD

Pricing Date03/19/1998Interest Accrual Date05/21/19981st Settle Date05/21/19981st Coupon Date06/21/1998

Identifiers

FIGIBBG00005ZYX9ISINZAG000016320ID NumberCP5073947Bond RatingsMoody'sBa2S&PBBFitchBB-CompositeBB-Issuance & TradingAmt Issued/OutstandingZAR378,175,000.00 (M) /ZAR378,175,000.00 (M)Min Piece/Increment1.00/ 1.00Par Amount1.00Book RunnerExchangeJSE Contrib Prx

Australia 61 2 9777 8600Japan 81 3 4565 8900Brazil 5511 2395 9000Singapore 65 6212 1000Europe 44 20 7330 7500U.S. 1 212 318 2000Germany 49 69 9204 1210Hong Kong 852 2977 6000Copyright 2022 Bloomberg Finance L.P.SN 519828 SAST GMT+2:00 G593-8248-173 14-Aug-2022 21:46:37

Additional information: The "books close dates" are 11 June and 11 December.

Required

- 1) A dealer is selling the bond presented in the snapshot above. Determine how much you will pay for this bond if the transaction date is today. Work with a par value of R1,000.
- 2) Show how the next coupon will be shared between the buyer and the seller.
- 3) Determine the clean price of the bond of today.
- 4) Determine the bond clean price assuming the bond is trading ex-coupon (assume settlement date is 7 days before next coupon date).

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SAGB 10 1/2 12/21/26 #	18.580	+0.048	8.598 / 8.562	-1/2 p.a. ; BEV
As of	12 A			Source B
SAGB 10 1/2 12/21/26 #	Actions	Settings	Page 1/12	Security Description: Bond
94	No Notes	95 Buy	96 Sell	
29 Bond Description	26 Issuer Description	Identifiers		
10 Bond Info	Name REPUBLIC OF SOUTH AFRICA	FIGI	BBG00005ZYX9	
12 Addtl Info	Industry Treasury (BCLASS)	ISIN	ZAG000016320	
13 Reg/Tax	Security Information	ID Number	CP5073947	
14 Covenants	Mkt Iss DOMESTIC	Bond Ratings		
15 Guarantors	Ctry/Reg ZA	Currency ZAR	Moody's Ba2	
16 Bond Ratings	Rank Sr Unsecured	Series R186	S&P BB	
17 Identifiers	Coupon 10.500000	Type Fixed	Fitch IBCA BB-	
18 Exchanges	Cpn Freq S/A	semi annual	Composite BB-	
19 Inv Parties	Day Cnt ACT/365	Iss Price 77.36654	Issuance & Trading	
20 Fees, Restrict	Maturity 12/21/2026		Amt Issued/Outstanding	
21 Schedules	BULLET		ZAR 378,175,000.00 (M)	
22 Coupons	Iss Sprd		ZAR 378,175,000.00 (M)	
Quick Links	Calc Type (368)S AFRICA EX-DIV BD	Min Piece/Increment		
32 ALLQ Pricing	Pricing Date 03/19/1998	1.00/ 1.00		
33 QRD Qt Recap	Interest Accrual Date 05/21/1998	Par Amount 1.00		
34 TDH Trade Hist	1st Settle Date 05/21/1998	Book Runner		
35 CACS Corp Action	1st Coupon Date 06/21/1998	Exchange	JSE Contrib Prx	
36 CF Prospectus				
37 CN Sec News				
38 HDS Holders				
60 Send Bond				

Additional information: The "books close dates" are 11 June and 11 December.

Required

- 1) A dealer is selling the bond presented in the snapshot above. Determine how much you will pay for this bond if the transaction date is today. Work with a par value of R1,000.

2/10/25

- 2) Show how the next coupon will be shared between the buyer and the seller.
- 3) Determine the clean price of the bond of today.
- 4) Determine the bond clean price assuming the bond is trading ex-coupon (assume settlement date is 7 days before next coupon date).

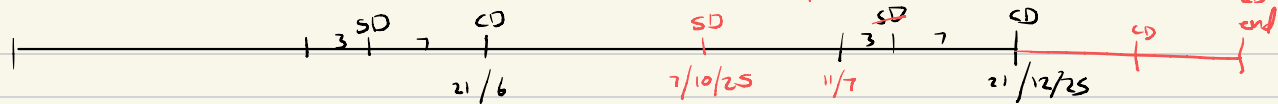
SD

12:00

$$SD = t+3$$

$$\therefore 2/10/25 + F + M + T = 7/10/25$$

1.



$$V_B = C \times \frac{1 - \left(\frac{1}{(1+r)^n}\right)}{r} + \frac{FV}{(1+r)^n}$$

$$C = 10,5\% \times 11 \div 2$$

$$= R52,5$$

value @ NCD (Disc FCFs)

$$r = \frac{8,562\%}{2} = 4,281\%$$

$$V = 52,5 \times \frac{1 - \left(\frac{1}{(1+5,75\%)^{1 \times 2}}\right)}{5,75\%} + \frac{1000}{(1+5,75\%)^{1 \times 2}} + 52,5$$

$$= R1000,83 \times 1070,70$$

Discount to today ^{SD} ×

$$\frac{FV}{(1+r)^n} = \frac{1000,83}{(1+5,75\%)^{\frac{80}{365}}}$$

$$AIP = \frac{P_{NCD}}{(1+r)^{\frac{d_1}{d_2}}}$$

$$= R988,64$$

$$1052,46$$

(dirty price)

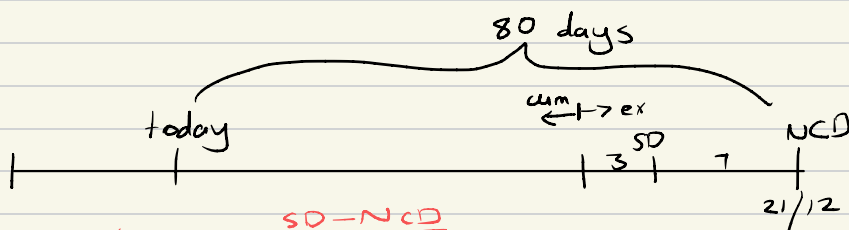
$$d_1 = SD - NCD$$

$$= 75$$

$$d_2 = LCD - NCD$$

$$= 183$$

2.



$$\text{Buyer's portion: } \frac{SD - NCD}{365} \times \text{Annual } C$$

$$= \frac{75}{365} \times 105 = 21,58$$

(based on NODs held bond)

$$\text{Seller's portion: } \frac{LCD - SD}{365} \times \text{Annual } C$$

$$= \frac{108}{365} \times 105 = 31,07$$

3.

$$\text{Clean price} = DP - AI$$

$$= R988,64 - 31,07$$

$$= R977,13$$

A number line diagram illustrating the interval $[11/12, 21/12]$. The interval is divided into three equal parts by tick marks at $11/12$, $15/12$, and $19/12$. The label "SD" is written above the middle tick mark, and the number "3" is written below it, indicating the number of subintervals.

$$d_1 = 1$$
$$AI = \frac{6}{365} \times 105$$

why SIP improvement

- Why const
- They want to see effects of prev cuts

Why not drop repo:

- ① Infl. is slightly above target (currently 3,5%/-)
They are worried infl will ↑ to 4% in next few months
(driven by electricity price increases)
- ② Stats are showing sign that economy is in good condition
(Q2 growth 0,8% - highest in 2 yrs)
they also provided some revised focus
∴ no urgent need to cut repo

Why not hike

- ① Didn't want to hurt economy which they believe still requires investments to grow
- ② Infl factor is no
- ③ Rand is also appreciation, meaning no urgent need to defend it

why was repo kept const — why not drop — give reason
- gives stats + be specific — why not hike — support with stats

0:11 Answer: infl 3,4 & worried still 1. cutting will exacerbate

QUESTION 2 (11marks)

John is watching updates from the Financial markets. He takes notice of 2 things; the opening yield of the R186 government bond, which matures in 2026, is at 8.15%. On the other hand Anglo bonds maturing in 2026 are trading at a yield of 11.4%. Both bond pay a coupon of 6%.

Required

1. John is confused, he doesn't understand why the yield of the Anglo bond is 3.25% (325basis points) higher than the yield of the R186 bond. Assist John with an explanation.
2. What will happen to the yield spread if South Africa slides into a recession.

QUESTION 3 (11marks)

You are given a Sappi bond which pays coupons on a semi-annual basis. The bond features are as follows; par value is R1,000, coupon rate is 7% and the bond has 12 semi-annual coupons remaining. Similar bonds are trading at a yield to maturity of 8%

The settlement period is t+3. The number of days from the valuation date (today) to the next coupon date is 86 days. There are 183 days in the coupon period and the bond is trading cum-coupon.

Required:

- a) How much will you pay for this bond? [7]
- b) Calculate the accrued interest [3]
- c) Calculate the clean price [1]

QUESTION 4

On the 14th November 2024, S&P Global Rating Agency revised its outlook on South African bonds from stable to positive. (Click this link in the footnote to read the article¹)

Required

1. What is the difference between a stable outlook and positive outlook. [2]
2. State and explain four reasons why S&P decided to revise its outlook on South Africa. [12]

¹ [https://amathuba.uct.ac.za/content/enforced/95004/S%20and%20P%20Ratings%20Credit%20Opinion%20-%2015%20November%202024%20\(1\).pdf](https://amathuba.uct.ac.za/content/enforced/95004/S%20and%20P%20Ratings%20Credit%20Opinion%20-%2015%20November%202024%20(1).pdf)

